ROSSMANN STORE SALES FORECASTING

An end-to-end retail forecasting workflow using exploratory analysis, Random Forest regression, LSTM deep learning and a Streamlit interface

PROJECT REPORT	PRIMARY FOCUS	SUBMISSION DATE
Final blog-style analysis	Deep Learning & LSTM	15 November 2026

Prepared from the supplied original assignment brief, five executed notebooks and dashboard / MLflow screenshots.

Evidence note: Local dashboard and MLflow model-version evidence were supplied. GitHub URL, public deployment URL and DVC multi-version screenshots were not supplied. These remain outstanding and are marked clearly in this report.

Executive Summary

The business objective is to forecast daily sales across Rossmann stores up to six weeks ahead, supporting finance planning and store-level decisions. The supplied assignment asks for customer-behaviour exploration, machine-learning and deep-learning sales prediction, and a web interface that displays predicted Sales and Customers. *Source: original Project 6 brief.*

The supplied work is organized into five notebooks: data merging and preprocessing; EDA; feature engineering and leakage-aware preprocessing; Random Forest modelling and evaluation; and LSTM time-series modelling with MLflow and Streamlit integration.

Workstream	Verified result from supplied notebook outputs
EDA	Sales–Customers correlation: Pearson 0.8236; Spearman 0.8319. Store-level promotion analysis: mean Sales uplift 41.43%.
Random Forest	Full-training RF: MAE 1,743.02; RMSE 2,367.11; RMSPE 41.54% (46,830 validation rows).
Sales LSTM	MAE 907.25; RMSE 1,257.80; MAPE 16.29%; RMSPE 21.59%.
Customers LSTM	MAE 60.0835; RMSE 95.814; MAPE 9.53%; RMSPE 12.82%.
Integrated app test	Store 1, seven dates: predicted Sales 18,720.35; predicted Customers 1,879; 21/21 programmatic checks passed.
Readiness	Targeted audit: 7/7 tracked sections and 26/26 artifact checks passed. Public deployment and final readiness remained pending.

The LSTM and Random Forest metrics come from different model setups and should not be treated as a controlled head-to-head comparison. Dashboard totals are sample outputs, not accuracy measures.

1. Business Context and Objectives

Rossmann operates multiple retail stores whose sales may vary with promotions, competition, school and state holidays, seasonality and locality. The assignment describes Sales as the prediction target, Customers as daily customer count, Store as store identifier and Open as the store operating indicator. The intended product is an end-to-end forecasting service for analysts in the finance team.

Project objectives

• Explore purchasing behaviour and interactions among Sales, Customers, promotions, holidays, store characteristics and competition.
• Prepare model-ready features with date-derived variables, missing-value handling, categorical encoding and numerical scaling.
• Develop a reproducible scikit-learn regression pipeline, inspect feature importance and estimate prediction intervals.
• Build store-wise LSTM forecasting with stationarity testing, differencing, ACF/PACF and sliding windows.
• Track the model with MLflow and serve Sales and Customers predictions through Streamlit.

2. Data Preparation and Feature Engineering

Notebook 03 records the chronological split: training 1 Jan 2013–19 Jun 2015; validation 20 Jun–31 Jul 2015 (42 days); and test 1 Aug–17 Sep 2015. The raw feature matrices contain 25 columns. There are 970,379 training rows, 46,830 validation rows and 41,088 test rows.

Partition	Rows	Raw features
Training	970,379	25
Validation	46,830	25
Test	41,088	25
Processed feature matrices	970,379 / 46,830 / 41,088	1,149 features

The final raw schema contained 20 numerical and 5 categorical features. The preprocessing pipeline expanded these to 1,149 processed features. The resulting train, validation and test matrices had zero missing values and zero infinite values. Notebook 03 records nine model-ready artifacts saved and verified, including processed matrices, targets, test IDs, feature names, metadata and the fitted preprocessor.

The notebook documents date/calendar, holiday-proximity, competition, promotion, store-level and categorical feature preparation. Median imputation was assigned to numerical fields and most-frequent imputation to categorical fields inside the preprocessing pipeline. StandardScaler and OneHotEncoder were fitted using X_train only; unseen categories were handled with OneHotEncoder(handle_unknown='ignore').

Missing competition metadata was investigated at store level. CompetitionDistance was available for 1,112 of 1,115 stores; Stores 291, 622 and 879 had consistently missing values. No missing-to-available transition was observed in the training period. Missing distance must not be interpreted as proof that no competitor existed.

3. Exploratory Data Analysis and Business Insights

The EDA notebook covers data quality, promotion distributions, temporal patterns, holiday behaviour, customer traffic, store types, assortment, competition and correlations. These are descriptive analyses: they identify patterns but do not establish causation.

Promotions and customer traffic

Across 844,392 valid open-store records and 1,115 stores, the paired store-level analysis found mean Sales increasing from 5,906.57 without promotions to 8,205.18 during promotions. Mean store-level Sales uplift was 41.43% (median 40.56%). Average store-level Customers increased from 686.90 to 838.29, a mean uplift of 23.84% (median 22.66%). Positive Sales uplift was observed in 1,114 of 1,115 stores; positive customer uplift in 1,101 stores.

Promotion uplift varied by StoreType. Sales uplift: Type A +42.96%, B +18.20%, C +33.41%, D +36.93%. Customer uplift: Type A +25.53%, B +10.80%, C +16.66%, D +20.12%. This supports examining Promo together with store characteristics, but does not account for promotional cost or profit margin.

Sales, Customers and correlation

Customers had the strongest positive individual association with Sales among the variables discussed: Pearson +0.8236 and Spearman +0.8319. Promo was positively associated with Sales (Pearson +0.3681; Spearman +0.4204). Promo2 showed a weak negative association (Pearson –0.1276; Spearman –0.1166). SchoolHoliday and CompetitionDistance had very weak individual correlations. Historical Customers is not available as an actual future input in the test data, so it cannot be used as a known future Sales predictor.

Seasonality, store type and competition

December recorded the highest mean Sales in both 2013 (8,613.46) and 2014 (8,603.81). Store Type B had the highest mean Sales (10,231.41) and Customers (2,021.82), but represented only 17 stores. Assortment B mean Sales was 8,639.35, but represented only nine stores. Small groups and one-store combinations should not be generalized.

CompetitionDistance had a very weak negative association with average Sales (Pearson –0.0424; Spearman –0.0280). For the competitor-opening comparison, 188 stores had eligible observations in both 12-month windows. Mean Sales decreased 4.41% after the recorded opening date and Customers decreased 4.92%. Opening month/year was approximated using the first day of the month; this is not a causal estimate and no separate reopening date was available.

EDA implications

• Promo, calendar effects, store type and store identity are useful candidates for feature engineering.
• Customers is strongly associated with historical Sales, but actual future Customers are unavailable and must not leak into Sales forecasts.
• CompetitionDistance alone is a weak descriptive signal.
• Small sample groups and descriptive comparisons require cautious interpretation.

4. Machine-Learning Modelling: Random Forest

Notebook 04 compares DummyRegressor baselines with a Random Forest regression pipeline, performs controlled tuning, summarizes original-feature importance, estimates tree-based prediction intervals and serializes timestamped model bundles with metadata.

Model	Training rows	MAE	RMSE	RMSPE
Dummy – full training	970,379	2,757.65	3,733.83	45.78%
Dummy – pilot	60,000	2,758.12	3,734.57	45.69%
Selected RF pilot	60,000	1,898.08	2,570.83	42.39%
RF full training	970,379	1,743.02	2,367.11	41.54%

The full-training RF reduced error versus the full-training Dummy baseline by 36.79% in MAE, 36.60% in RMSE and 9.25% in RMSPE. The full-training model used 60 trees, max_depth 18, min_samples_split 20, min_samples_leaf 10, max_features sqrt, bootstrap enabled, max_samples 0.8, squared_error criterion and random_state 42. It was fitted on 970,379 rows and evaluated on 46,830 validation rows.

Top original-feature importance in the full-training model

Rank	Feature	Importance
1	Open	23.627%
2	DayOfWeek	21.603%
3	Promo	15.167%
4	Store	10.487%
5	IsWeekend	9.603%
6	StateHoliday	8.799%
7	DaysUntilHoliday	1.559%
8	StoreType	1.203%
9	Assortment	1.140%
10	WeekOfYear	1.078%

Prediction intervals and caveat

The Random Forest interval method used the distribution of tree predictions as an empirical interval estimate. On full-training validation, nominal coverage was 90%, while observed coverage was 63.48% overall and 57.54% on positive-Sales observations. These intervals were not calibrated to their nominal coverage and should not be described as reliable 90% confidence intervals. The notebook verifies that the timestamped model bundle, metadata and preprocessor reload successfully and that predictions match after reload.

Validation metrics were used during model selection and are not an independent final test estimate. No test-target accuracy is claimed here.

5. Deep Learning: Store-Wise LSTM Forecasting

The assignment asks for a Long Short-Term Memory recurrent neural network and recommends a compact model that can run in Google Colab. Notebook 05 follows a store-wise time-series workflow: missing-date audit, Augmented Dickey–Fuller stationarity testing, conditional differencing, ACF/PACF analysis, supervised sliding windows, training-only MinMax scaling in the (–1, 1) range and chronological validation.

Time-series preparation

- Prepare individual store-wise Sales and Customers sequences.
- Check calendar continuity and investigate missing dates rather than treating absent dates as zero sales.
- Use the Augmented Dickey–Fuller test to assess stationarity and difference non-stationary series where indicated.
- Inspect autocorrelation and partial autocorrelation to understand temporal dependence.
- Create supervised learning samples using a 42-day historical lookback to predict the next day.
- Fit MinMax scaling on training-period data only; validation starts 20 June 2015.

Sales LSTM training and evaluation

The Sales model trained for 12 epochs and restored the best weights from epoch 11. Best scaled validation MSE was 0.0045324 and best scaled validation MAE was 0.0507471. Training took approximately 8.64 minutes. Validation contained 46,830 observations representing 1,115 stores over 42 dates.

Sales LSTM metric	Validation result
MAE	907.25
RMSE	1,257.80
MAPE (positive actual Sales only)	16.29%
RMSPE (positive actual Sales only)	21.59%
Validation observations	46,830
Zero actual Sales excluded from percentage metrics	6,548

Percentage metrics exclude zero actual Sales rows. The Sales LSTM uses historical Sales as its univariate input. Holiday, weekend and promotion fields are accepted and validated by the dashboard but are not direct inputs to this LSTM.

Customers LSTM extension

A separate Customers LSTM was trained to satisfy the dashboard requirement to show customer-number predictions. It uses historical Customers only, a 42-day lookback, training-only MinMax scaling in (–1, 1), chronological validation beginning 20 June 2015 and Huber loss. The recorded architecture contains one LSTM layer followed by Dense layers, with 4,897 trainable parameters. It trained for 12 epochs; best validation loss was at epoch 12.

Customers LSTM metric	Validation result
MAE	60.0835
RMSE	95.814
MAPE (positive actual Customers only)	9.53%
RMSPE (positive actual Customers only)	12.82%
Training windows	278,750
Validation windows	46,830
Recursive test forecasts	41,088 rows (856 stores × 48 dates)

Sales and Customers errors are measured in different units and scales; their MAE/RMSE values are not directly comparable.

6. Forecast Uncertainty, MLflow and Model Governance

Sales LSTM uncertainty

The Sales LSTM notebook estimates uncertainty using absolute validation residuals. The residuals were split chronologically into 29 calibration dates (20 June–18 July 2015) and 13 held-out evaluation dates (19–31 July 2015). The 95th percentile of absolute calibration residuals was 2,232.38. Held-out nominal 95% interval coverage was 89.13%, with average interval width 3,858.38 and median width 3,418.38. These are empirical residual-based intervals, not guaranteed 95% confidence intervals.

Lag sensitivity analysis

Notebook 05 includes a lag sensitivity analysis with 42 lag records, a CSV, a plot and metadata. This is a model-response diagnostic, not causal feature importance.

MLflow tracking

The LSTM workflow uses MLflow 3.18.0, a local SQLite tracking URI and experiment `Project6_Rossmann_LSTM_Forecasting`. The notebook logs the Sales model, reloads it from MLflow and runs inference on 41,088 test rows. Maximum and mean prediction differences between the MLflow-loaded model and the Section 9 forecast were both 0.0. The supplied screenshot shows two completed LSTM experiment runs.

Random Forest model registry

The supplied MLflow screenshots show registered model `Rossmann_RandomForest_Sales` with two READY versions. Version 1 is linked to `RandomForest_Pilot`; Version 2 is linked to `RandomForest_FullTraining`. This represents a pilot-to-full-training progression. These screenshots do not establish DVC data-version history.

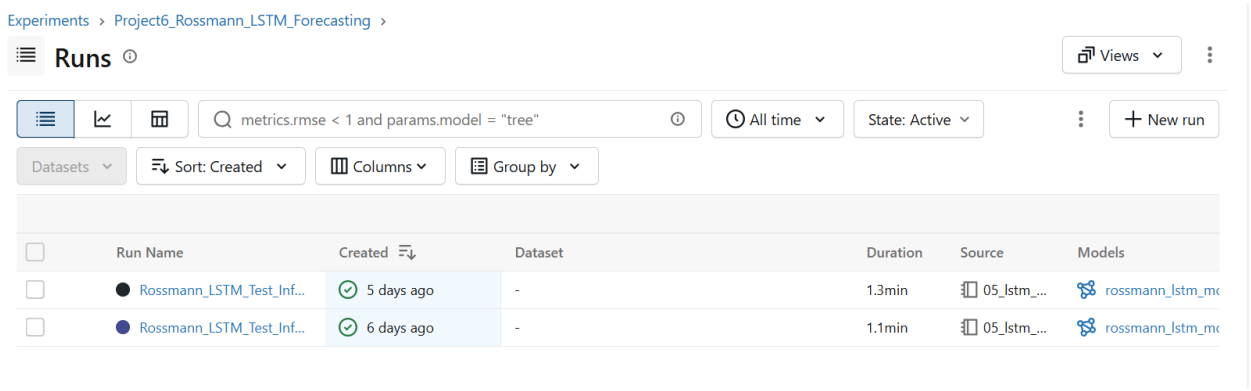


Figure 1. MLflow LSTM experiment with two completed runs.

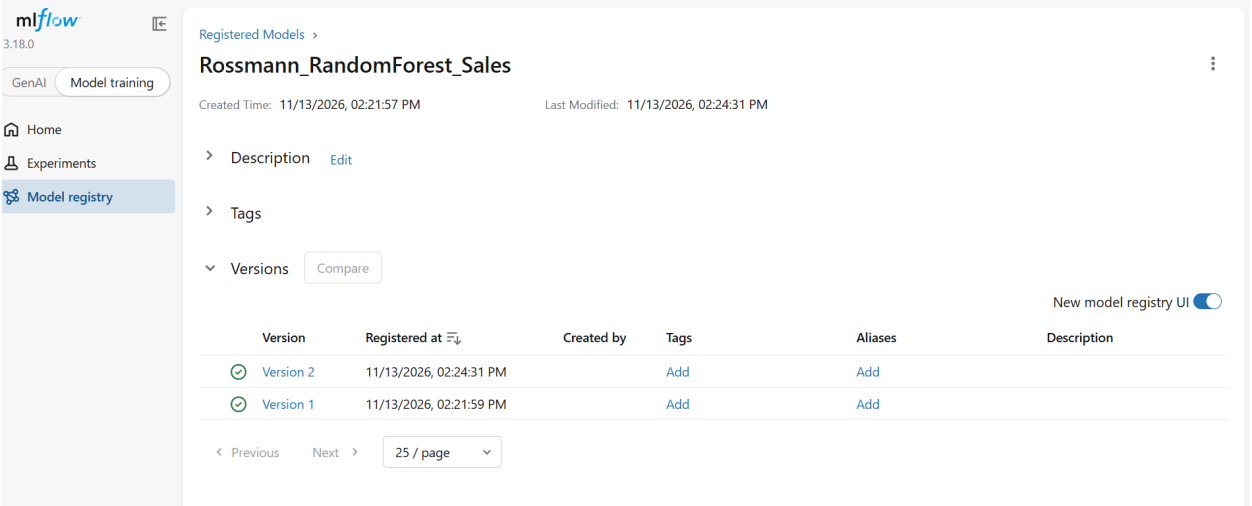


Figure 2. MLflow registry showing two Random Forest model versions.

7. Streamlit Dashboard and End-to-End Testing

The Streamlit interface provides a Store ID selector and future-date CSV upload. Date is required; IsHoliday, IsWeekend, IsPromo and Open are optional inputs. The integrated application loads the saved Sales LSTM, Customers LSTM, Customers scaler and preprocessing bundle. It provides Sales and Customers predictions, visualizations and combined CSV download.

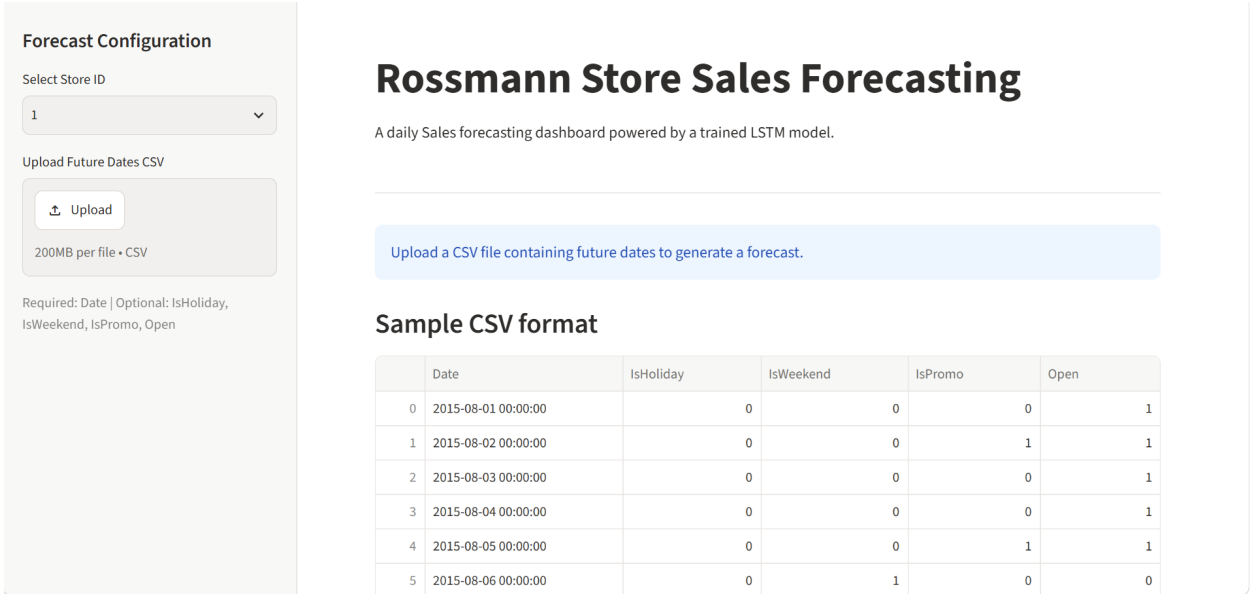


Figure 3. Dashboard landing page and future-date CSV upload.

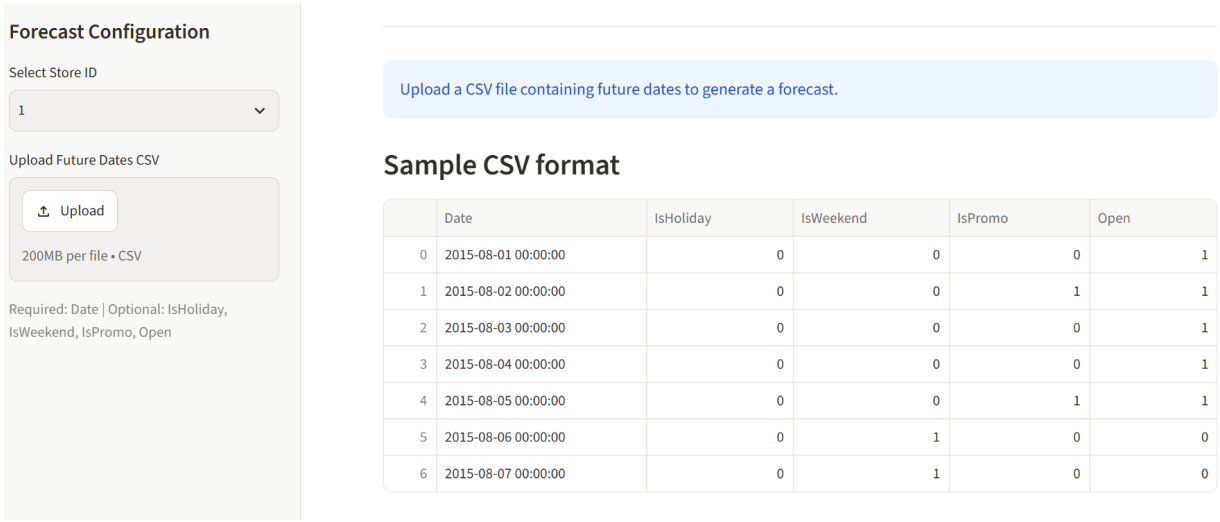


Figure 4. Forecast summary and daily prediction table.

Programmatic end-to-end test

Section 17 tested Store ID 1 for seven dates, 18–24 September 2015. It generated seven aligned Sales and Customers rows and verified output columns, finite/non-negative predictions, closed-store zero handling, interval-bound ordering and CSV/JSON report saving. All 21/21 checks passed. The sample totals were 18,720.35 predicted Sales and 1,879 predicted Customers. These totals are sample output values, not model accuracy metrics.

Manual browser testing

Section 14.9 documents local browser testing for Store IDs 1 and 3 using a seven-day sample CSV. Both forecast targets and visualizations were displayed, and combined CSV files were downloaded. Store 1 totals: 18,720.35 Sales and 1,879 Customers. Store 3 totals: 26,660.02 Sales and 2,673 Customers. The tests verify local functionality, not public deployment or full-horizon forecast accuracy.

8. Diagnostics, Limitations and Readiness

Open-day zero-prediction investigation

For 19 September 2015, the notebook found negative inverse-scaled predictions on dates where Open = 1. These were clipped to zero to maintain non-negative outputs. The immediate cause was negative model outputs followed by non-negative clipping, not the closed-store business rule. The notebook does not establish why the models generated those negative values and does not introduce an artificial positive floor.

Store	Target	Open	Inverse-scaled prediction	Final prediction
1	Sales	1	85.9273	85.9273
1	Customers	1	–96.9125	0
3	Sales	1	–1,162.9421	0
3	Customers	1	–88.7143	0

Limitations to disclose

• Sales and Customers LSTMs are univariate and use historical target values only; holiday, weekend and promotion variables are not direct inputs.
• Empirical prediction intervals are not guaranteed 95% confidence intervals. Random Forest interval coverage was below its nominal 90% level.
• Open-day zero predictions have an identified immediate cause, but the underlying model behaviour remains unexplained.
• The dashboard was tested locally; local execution does not establish public hosting.
• Notebook 05 Section 18 is a targeted audit of selected sections and artifacts, not a fresh re-execution of every notebook cell.

Final readiness audit

Notebook 05 Section 18 reports seven tracked sections passed (10.7, 12, 13, 15, 16, 17 and 17.4) and 26/26 artifact checks passed. It explicitly records Public deployment: PENDING and Final submission readiness: PENDING. The report therefore does not claim that all submission requirements are complete.

9. Conclusion and Submission Checklist

The supplied work implements a substantial forecasting workflow: EDA, leakage-aware preprocessing, Dummy and Random Forest regression, Sales LSTM, Customers LSTM, MLflow tracking and a Streamlit interface. The most complete deployment evidence is the integrated local workflow: both forecast targets can be generated for selected stores, with saved artifacts and programmatic checks documenting the process. The report also preserves the important limitations around univariate inputs, imperfect interval coverage and negative inverse-scaled predictions.

Assignment requirement	Evidence/status based on supplied materials
Blog-style PDF with Deep Learning emphasis	This report draft prepared from supplied notebooks and evidence.
GitHub code link including notebooks	Public URL not supplied; insert the repository link.
Deployed app link and/or dashboard screenshot	Local dashboard screenshots supplied; public deployment is pending in the final audit.
Multiple data versions in DVC store	DVC multi-version screenshots not supplied; add before submission.
Multiple model versions in MLflow	Supplied: Random Forest registry shows Version 1 and Version 2; LSTM runs screenshot supplied.
Other supporting evidence	Dashboard screenshots, LSTM runs and Random Forest registry screenshots supplied.

Links to add before final submission

GitHub repository: [Insert public repository URL]
Deployed application: [Insert public app URL, if hosted]
DVC evidence: [Add screenshot(s) showing multiple data versions]

This document is a content-complete report draft based on the files provided, but it is not yet a fully submission-ready package until the missing repository/deployment links and DVC evidence are attached or confirmed. The assignment brief requires a GitHub link, deployed app link and/or dashboard screenshot, and other evidence screenshots.

End of Report